

```
Layer2Switch-1>enable
```

```
Layer2Switch-1#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Layer2Switch-1(config)#vlan 20
```

```
Layer2Switch-1(config-vlan)#vlan 333
```

```
Layer2Switch-1(config-vlan)#exit
```

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface range g0/0-1
```

```
Layer2Switch-1(config-if-range)#channel-group 1 mode active
```

Creating a port-channel interface Port-channel 1

```
Layer2Switch-1(config-if-range)#exit
```

```
Layer2Switch-1(config)#interface port-channel 1
```

```
Layer2Switch-1(config-if)#switchport mode trunk
```

Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.

```
Layer2Switch-1(config-if)#switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)#exit
```

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface range g0/2-3
```

```
Layer2Switch-1(config-if-range)#channel-group 2 mode active
```

Creating a port-channel interface Port-channel 2

```
Layer2Switch-1(config-if-range)#exit
```

```
Layer2Switch-1(config)#interface port-channel 2
```

```
Layer2Switch-1(config-if)#switchport mode trunk
```

Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.

```
Layer2Switch-1(config-if)#switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)#exit
```

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface range g1/0-1
```

```
Layer2Switch-1(config-if-range)#channel-group 3 mode active
```

Creating a port-channel interface Port-channel 3

```
Layer2Switch-1(config-if-range)#exit
```

```
Layer2Switch-1(config)#interface port-channel 3
```

```
Layer2Switch-1(config-if)#switchport mode trunk
```

Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.

```
Layer2Switch-1(config-if)#switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)#exit
```

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface range g1/2-3
```

```
Layer2Switch-1(config-if-range)#channel-group 4 mode active
```

Creating a port-channel interface Port-channel 4

```
Layer2Switch-1(config-if-range)#exit
```

```
Layer2Switch-1(config)#interface port-channel 4
```

```
Layer2Switch-1(config-if)#switchport mode trunk
```

Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.

```
Layer2Switch-1(config-if)#switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)#exit
```

```
ayer2Switch-1#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface port-channel 1
```

```
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
```

```
Layer2Switch-1(config-if)# switchport mode trunk
```

```
Layer2Switch-1(config-if)# switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)# exit
```

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface port-channel 2
```

```
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
```

```
Layer2Switch-1(config-if)# switchport mode trunk
```

```
Layer2Switch-1(config-if)# switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)# exit
```

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface port-channel 3
```

```
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
```

```
Layer2Switch-1(config-if)# switchport mode trunk
```

```
Layer2Switch-1(config-if)# switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)# exit
```

```
Layer2Switch-1(config)#
```

```
Layer2Switch-1(config)#interface port-channel 4
```

```
Layer2Switch-1(config-if)# switchport trunk encapsulation dot1q
```

```
Layer2Switch-1(config-if)# switchport mode trunk
```

```
Layer2Switch-1(config-if)# switchport trunk native vlan 333
```

```
Layer2Switch-1(config-if)# exit
```

```
Layer2Switch-1(config)#
```

2

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#interface range g0/0-1

Switch(config-if-range)#channel-group 1 mode active

Creating a port-channel interface Port-channel 1

Switch(config-if-range)#exit

Switch(config)#interface port-channel 1

Switch(config-if)#switchport trunk encapsulation dot1q

Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 333

Switch(config-if)#exit

Switch(config)#interface range g0/2-3

Switch(config-if-range)#channel-group 2 mode active

Creating a port-channel interface Port-channel 2

Switch(config-if-range)#exit

Switch(config)#interface port-channel 2

Switch(config-if)#switchport trunk encapsulation dot1q

Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 333

Switch(config-if)#exit

Switch(config)#interface range g1/0-1

Switch(config-if-range)#channel-group 3 mode active

Creating a port-channel interface Port-channel 3

Switch(config-if-range)#exit

```
Switch(config)#interface port-channel 3
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#exit
Switch(config)#interface range g1/2-3
Switch(config-if-range)#channel-group 4 mode active
Creating a port-channel interface Port-channel 4
```

```
Switch(config-if-range)#exit
Switch(config)#interface port-channel 4
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#exit
Switch(config)#
```

3

enable

configure terminal

vlan 20

vlan 333

exit

```
interface range g0/0-1
channel-group 1 mode active
exit
interface port-channel 1
switchport mode trunk
switchport trunk native vlan 333
exit
```

```
interface range g0/2-3
channel-group 2 mode active
exit
interface port-channel 2
switchport mode trunk
switchport trunk native vlan 333
exit
```

```
default interface g1/0
interface g1/0
switchport mode access
switchport access vlan 20
exit
```

```
interface g1/1
switchport mode access
switchport access vlan 333
exit
```

```
Switch>enable
```

```
Switch#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#
```

```
Switch(config)#interface port-channel 1
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

Command rejected: Po1 is not a switching port.

```
Switch(config-if)# switchport mode trunk
```

Command rejected: Po1 is not a switching port.

```
Switch(config-if)# switchport trunk native vlan 333
```

Command rejected: Po1 is not a switching port.

```
Switch(config-if)# exit
```

```
Switch(config)#
```

```
Switch(config)#interface port-channel 2
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

Command rejected: Po2 is not a switching port.

```
Switch(config-if)# switchport mode trunk
```

Command rejected: Po2 is not a switching port.

```
Switch(config-if)# switchport trunk native vlan 333
```

Command rejected: Po2 is not a switching port.

```
Switch(config-if)# exit
```

```
Switch(config)#
```

```
Switch#enable
```

```
Switch#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#
```

```
Switch(config)#no interface port-channel 1
```

```
Switch(config)#no interface port-channel 2
```

```
Switch(config)#
```

```
Switch(config)#interface range g0/0-1
```

```
Switch(config-if-range)# channel-group 1 mode active
```

Creating a port-channel interface Port-channel 1

```
Switch(config-if-range)# exit
```

```
Switch(config)#interface port-channel 1
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# switchport trunk native vlan 333
```

```
Switch(config-if)# exit
```

```
Switch(config)#
```

```
Switch(config)#interface range g0/2-3
```

```
Switch(config-if-range)# channel-group 2 mode active
```

Creating a port-channel interface Port-channel 2

```
Switch(config-if-range)# exit
```

```
Switch(config)#interface port-channel 2
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# switchport trunk native vlan 333
```

```
Switch(config-if)# exit
```

```
Switch#show etherchannel summary
```

Flags: D - down P - bundled in port-channel

 I - stand-alone s - suspended

 H - Hot-standby (LACP only)

 R - Layer3 S - Layer2

 U - in use f - failed to allocate aggregator

 M - not in use, minimum links not met

 u - unsuitable for bundling

w - waiting to be aggregated

d - default port

Number of channel-groups in use: 2

Number of aggregators: 2

Group Port-channel Protocol Ports

1	Po1(SU)	LACP	Gi0/0(P)	Gi0/1(P)
2	Po2(SU)	LACP	Gi0/2(P)	Gi0/3(P)

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Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#

Switch(config)#default interface g1/0

Interface GigabitEthernet1/0 set to default configuration

Switch(config)#default interface g1/1

Interface GigabitEthernet1/1 set to default configuration

Switch(config)#

Switch(config)#interface g1/0

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 20

Switch(config-if)# exit

```
Switch(config)#  
Switch(config)#interface g1/1  
Switch(config-if)# switchport mode access  
Switch(config-if)# switchport access vlan 333  
Switch(config-if)# exit  
Switch(config)#
```

```
Switch#enable  
Switch#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#  
Switch(config)#interface range g0/0-1  
Switch(config-if-range)# channel-group 1 mode active  
Creating a port-channel interface Port-channel 1
```

```
Switch(config-if-range)# exit  
Switch(config)#interface port-channel 1  
Switch(config-if)# switchport trunk encapsulation dot1q  
Switch(config-if)# switchport mode trunk  
Switch(config-if)# switchport trunk native vlan 333  
Switch(config-if)# exit  
Switch(config)#  
Switch(config)#interface range g0/2-3  
Switch(config-if-range)# channel-group 2 mode active  
Creating a port-channel interface Port-channel 2  
  
Switch(config-if-range)# exit  
Switch(config)#interface port-channel 2
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# switchport trunk native vlan 333
```

```
Switch(config-if)# exit
```

5

```
Switch#enable
```

```
Switch#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#
```

```
Switch(config)#hostname Layer2Switch-5
```

```
Layer2Switch-5(config)#
```

```
Layer2Switch-5(config)#vlan 20
```

```
Layer2Switch-5(config-vlan)#vlan 333
```

```
Layer2Switch-5(config-vlan)#exit
```

```
Layer2Switch-5(config)#
```

```
Layer2Switch-5(config)#default interface g1/0
```

Interface GigabitEthernet1/0 set to default configuration

```
Layer2Switch-5(config)#default interface g1/1
```

Interface GigabitEthernet1/1 set to default configuration

```
Layer2Switch-5(config)#
```

```
Layer2Switch-5(config)#interface g1/0
```

```
Layer2Switch-5(config-if)# switchport mode access
```

```
Layer2Switch-5(config-if)# switchport access vlan 20
```

```
Layer2Switch-5(config-if)# exit
```

```
Layer2Switch-5(config)#
```

```
Layer2Switch-5(config)#interface g1/1
```

```
Layer2Switch-5(config-if)# switchport mode access  
Layer2Switch-5(config-if)# switchport access vlan 333  
Layer2Switch-5(config-if)# exit  
Layer2Switch-5(config)#
```

```
Layer2Switch-5>enable  
Layer2Switch-5#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Layer2Switch-5(config)#interface range g0/0-1  
Layer2Switch-5(config-if-range)#channel-group 1 mode active  
Creating a port-channel interface Port-channel 1
```

```
Layer2Switch-5(config-if-range)#exit  
Layer2Switch-5(config)#interface port-channel 1  
Layer2Switch-5(config-if)#switchport trunk encapsulation dot1q  
Layer2Switch-5(config-if)#switchport mode trunk  
Layer2Switch-5(config-if)#switchport trunk native vlan 333  
Layer2Switch-5(config-if)#exit  
Layer2Switch-5(config)#  
Layer2Switch-5(config)#interface range g0/2-3  
Layer2Switch-5(config-if-range)#channel-group 2 mode active  
Creating a port-channel interface Port-channel 2
```

```
Layer2Switch-5(config-if-range)#exit  
Layer2Switch-5(config)#interface port-channel 2  
Layer2Switch-5(config-if)#switchport trunk encapsulation dot1q  
Layer2Switch-5(config-if)#switchport mode trunk  
Layer2Switch-5(config-if)#switchport trunk native vlan 333
```

```
Layer2Switch-5(config-if)#exit
```

```
Layer2Switch-5(config)#
```

```
ping 192.168.20.3
```

```
84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=15.326 ms
```

```
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=4.866 ms
```

```
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=11.776 ms
```

```
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=12.157 ms
```

```
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=8.337 ms
```

```
PC1>
```

```
PC1> ping 192.168.20.5
```

```
84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=9.502 ms
```

```
84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=5.597 ms
```

```
84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=8.269 ms
```

```
84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=11.700 ms
```

```
84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=23.659 ms
```

```
PC1>
```

```
ping 192.168.33.2
```

```
No gateway found **разные VLAN**
```

PC1>

ping 192.168.33.4

84 bytes from 192.168.33.4 icmp_seq=1 ttl=64 time=21.761 ms

84 bytes from 192.168.33.4 icmp_seq=2 ttl=64 time=14.691 ms

84 bytes from 192.168.33.4 icmp_seq=3 ttl=64 time=5.729 ms

84 bytes from 192.168.33.4 icmp_seq=4 ttl=64 time=7.450 ms

84 bytes from 192.168.33.4 icmp_seq=5 ttl=64 time=11.802 ms

PC2> ping 192.168.33.6

84 bytes from 192.168.33.6 icmp_seq=1 ttl=64 time=16.505 ms

84 bytes from 192.168.33.6 icmp_seq=2 ttl=64 time=12.147 ms

84 bytes from 192.168.33.6 icmp_seq=3 ttl=64 time=21.698 ms

84 bytes from 192.168.33.6 icmp_seq=4 ttl=64 time=17.158 ms

84 bytes from 192.168.33.6 icmp_seq=5 ttl=64 time=12.400 ms

PC2>

ping 192.168.20.5

84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=7.263 ms

84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=11.499 ms

84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=20.998 ms

84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=24.797 ms

84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=9.145 ms

PC3> ping 192.168.33.6

No gateway found

PC3>

PC4> ping 192.168.33.6

84 bytes from 192.168.33.6 icmp_seq=1 ttl=64 time=13.538 ms

84 bytes from 192.168.33.6 icmp_seq=2 ttl=64 time=11.475 ms

84 bytes from 192.168.33.6 icmp_seq=3 ttl=64 time=21.406 ms

84 bytes from 192.168.33.6 icmp_seq=4 ttl=64 time=18.008 ms

84 bytes from 192.168.33.6 icmp_seq=5 ttl=64 time=14.707 ms

PC4>

ping 192.168.33.2

No gateway found

PC5>

Frame 21: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)

Destination: Private_66:68:02 (00:50:79:66:68:02)

Source: Private_66:68:00 (00:50:79:66:68:00)

Type: 802.1Q Virtual LAN (0x8100)

[Stream index: 5]

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20

Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.3

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x2579 (9593)

000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0xabdb [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.20.1

Destination Address: 192.168.20.3

[Stream index: 0]

Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0xa6e5 [correct]

[Checksum Status: Good]

Identifier (BE): 31013 (0x7925)

Identifier (LE): 9593 (0x2579)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Response frame: 22]

Data (56 bytes)

Frame 58: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:03 (00:50:79:66:68:03)

Destination: Private_66:68:03 (00:50:79:66:68:03)

Source: Private_66:68:01 (00:50:79:66:68:01)

Type: IPv4 (0x0800)

[Stream index: 7]

Internet Protocol Version 4, Src: 192.168.33.2, Dst: 192.168.33.4

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x2586 (9606)

000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x91cc [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.33.2

Destination Address: 192.168.33.4

[Stream index: 1]

Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0x99e5 [correct]

[Checksum Status: Good]

Identifier (BE): 34341 (0x8625)

Identifier (LE): 9606 (0x2586)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Response frame: 59]

Data (56 bytes)